

ASYMPTOTIC NORMALITY FOR A SMOOTH KERNEL ESTIMATOR OF THE CONDITIONAL QUANTILE FOR CENSORED TIME SERIES

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Summary: In this paper we study a smooth estimator of the conditional quantile function in the censorship model, when the data exhibit some dependence structure. We show that, under some regularity conditions, the kernel estimator of the conditional quantile suitably normalised is asymptotically normally distributed. An application to prediction and confidence intervals is also given. Some simulations have been drawn to lend further support to our theoretical results of the normality for finite samples sizes.

1. Introduction

In recent years, there is a resurgence in the theory of nonparametric estimation of conditional median and other quantiles (Berlinet et al., 2001; Hall et al.,

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2002; , Gannoun et al., 2005; Ould Saïd, 2006; Ould Saïd & Sadki, 2008a). Conditional quantiles have been shown to be very useful to make prediction with the robustness property especially in presence of outliers or when the distribution is heavy-tailed or asymmetric.

Concerning the censored case, Beran (1981) introduced a nonparametric estimate of the conditional survival function and proved some consistency results which were later exposed and extended by Dabrowska (1987, 1989). Further results, including bootstrap approximations have been obtained by Van Keilegom & Veraverbeke (1996, 1997, 1998). We consider censoring and dependent data: since we cannot get away from this type of data in many practical applications when event times are correlated. For example observations on duration of unemployment may be right censored and typically correlated. Our interest in this paper, is to extend our result (see Ould Saïd & Sadki, 2008a) in the iid case to the strong mixing one.

The rest of the paper is organised as follows. In Section 2, we recall the model and the definition of the estimators. The assumptions and main results are given in Section 3. Some applications are given in Section 4. Section 5 is devoted to some simulations to lend further support to our theoretical results. The proofs with some auxiliary results are relegated to Section 6.

2. Notations and definition of the estimator

Consider a sequence of strictly stationary rv's $T_1, T_2, \dots$ with common unknown absolutely continuous distribution function (df) F . In survival analysis, the random variables can be the lifetime of patients under study. In reality it is not possible to observe the survival time of all patients, and often some of them are still alive at the end of the study, or withdraw alive during the

study, or die from other causes than those under study. In this case, we observe another rv C called censoring time. Then the observable rv's are the minimum of the survival time and a censoring time C and an indicator telling whether the observation is censored or not. That is, assuming that $C_1, C_2, \dots, C_n$ are n censoring rv's with common unknown df G , we observe only the n pairs $\{(Y_i, \delta_i), i = 1, 2, \dots, n\}$, where $Y_i = T_i \wedge C_i$ and $\delta_i = \mathbb{I}_{\{T_i \leq C_i\}}$ (where $\mathbb{I}_A$ denotes the indicator function of the set A).

Let X be a covariate real-valued rv and $F(\cdot|\cdot)$ be the conditional df of T given $X = x$, that is

$$F(t|x) = \mathbb{E} [\mathbb{I}_{\{T \leq t\}} | X = x], \quad (1)$$

which can be written as $F(t|x) = \frac{F_1(x, t)}{\ell(x)}$ where ℓ is the marginal density of X and $F_1(x, t) = \int_{-\infty}^t f(x, y) dy$ with $f(\cdot, \cdot)$ the density of (X, T) .

We observe $\{(Y_i, \delta_i, X_i), i = 1, 2, \dots, n\}$. We suppose that $\{C_i, i \geq 1\}$ and $\{(X_i, T_i), i \geq 1\}$ are independent. This condition is reasonable whenever the censoring is not associated with the characteristic of the individuals under study.

We also suppose that $\{T_i, i \geq 1\}$ and $\{C_i, i \geq 1\}$ are two independent sequences of stationary α -mixing rv's (see the definition below) with mixing coefficients $\alpha_1(n)$ and $\alpha_2(n)$ respectively. Cai (1998) showed that $\{Y_i, i \geq 1\}$ is strongly mixing with coefficient $\alpha(n) = 4 \max(\alpha_1(n), \alpha_2(n))$. Consequently, we suppose that $\{(Y_i, \delta_i, X_i), i \geq 1\}$ is strong mixing with a mixing coefficient equal to α . The strong mixing condition is well known to be weaker than many dependence conditions and many stochastic processes are known to be strong mixing. For all fixed $p \in [0, 1]$, the p -th conditional

quantile of F given $X = x$ is defined by

$$\xi_p(x) = \inf\{t \in \mathbb{R}_+ : F(t|x) \geq p\}. \quad (2)$$

We consider the estimation of the parameter $\xi_p(x)$ which satisfy

$$F(\xi_p(x)|x) = p \quad (3)$$

Here we provide a consistent and asymptotically normal conditional quantile estimator under the condition that C and (X, T) are independent as in the recent paper of Ould Saïd & Sadki (2008a) (see also Carbonez et al., 1995 and Kohler et al., 2002). This condition is reasonable whenever the censoring is independent of the characteristics of the individual under study. Throughout this paper we also assume that $\tau_F < \tau_G$, where $\tau_H = \sup\{t; H(t) < 1\}$ for any distribution function H .

Remark 1 Denote by W the df of Y and write $F(\{a\}) = F(a) - F(a^-)$ where $F(a^-) = \lim_{\varepsilon \rightarrow 0} F(a - \varepsilon)$. By independence we have $1 - W = (1 - F)(1 - G)$ which implies that $\tau_W = \tau_F \wedge \tau_G$. Now it is well-known that for known that for $\tau_F > \tau_G$ the df F can not be consistently estimated in general. If $\tau_G = \tau_F$, then F can be estimated and the Kaplan-Meier estimator $\hat{F}$ is strongly consistent if and only if $F(\{\tau_G\}) > 0$ but $1 - G(\{\tau_G\}) = 0$ (see Remark 3 in Stute and Wang 1993).

It is clear that an estimator of (2) will be obtained by estimating the conditional df (1), hence it suffices to estimate $F_1(x, t)$ and $\ell(x)$. The density $\ell(x)$ is not affected by the censoring and therefore it can be estimated consistently by the well-known kernel estimator.

It is well known that a linear estimator of the conditional df $F(t|x)$ is given by (see Ould Saïd, 2006)

$$\tilde{F}_n(t|x) = \sum_{i=1}^n W_{in}(x) \frac{\delta_i \mathbb{I}_{\{Y_i \leq t\}}}{\bar{G}(Y_i)}, \quad (4)$$

where $W_{in}(\cdot)$ are the well-known Nadaraya-Watson weights which are measurable functions of x and depending on the sample $X_1, X_2, \dots, X_n$ and $\bar{G}(\cdot) := 1 - G(\cdot)$.

Let us define a smooth estimate of (1) in the censored case (for more detail see Ould Saïd & Sadki, 2008a) by

$$\begin{aligned} \tilde{F}_n(t|x) &= \frac{\frac{1}{h_K} \sum_{i=1}^n \frac{\delta_i}{\bar{G}(Y_i)} K\left(\frac{x - X_i}{h_K}\right) H\left(\frac{t - Y_i}{h_H}\right)}{\frac{1}{nh_{n,K}} \sum_{i=1}^n K\left(\frac{x - X_i}{h_K}\right)} \\ &=: \frac{\tilde{F}_{1,n}(x, t)}{\ell_n(x)} \end{aligned} \quad (5)$$

with the convention $0/0 = 0$. K is a probability density function (so called kernel function), $h_{n,K} =: h_K, h_{n,H} =: h_H$ are sequence (so called bandwidth) of positive real numbers which goes to zero as n goes to infinity and $\ell_n(\cdot)$ is the well-known kernel estimator of $\ell(\cdot)$.

In practice $\bar{G}$ is usually unknown, hence it impossible to use the estimator (5). Then, we replace $\bar{G}$ by the Kaplan-Meier (1958) estimate G_n given by

$$\bar{G}_n(t) := 1 - G_n(t) = \begin{cases} \prod_{i=1}^n \left(1 - \frac{1 - \delta_{(i)}}{n - i + 1}\right)^{I_{\{Y_{(i)} \leq t\}}} & \text{if } t < Y_{(n)} \\ 0 & \text{if } t \geq Y_{(n)} \end{cases}$$

where $Y_{(1)} < Y_{(2)} < \dots < Y_{(n)}$ are the order statistics of (Y_i) and $\delta_{(i)}$ the indicator of non censoring associated to $Y_{(i)}$.

Therefore the feasible estimator of the conditional df $F(\cdot|x)$ is given by

$$\begin{aligned} F_n(t|x) &= \frac{\sum_{i=1}^n \frac{\delta_i}{\bar{G}_n(Y_i)} K\left(\frac{x - X_i}{h_K}\right) H\left(\frac{t - Y_i}{h_H}\right)}{\sum_{i=1}^n K\left(\frac{x - X_i}{h_K}\right)} \\ &=: \frac{F_{1,n}(t, x)}{\ell_n(x)} \end{aligned} \quad (6)$$

where

$$F_{1,n}(x, t) = \frac{1}{nh_K} \sum_{i=1}^n \frac{\delta_i}{\bar{G}_n(Y_i)} K\left(\frac{x - X_i}{h_K}\right) H\left(\frac{t - Y_i}{h_H}\right).$$

Then, a natural estimator of (2), is given by

$$\xi_{p,n}(x) = \inf \{t; F_n(t|x) \geq p\}, \quad (7)$$

which satisfy

$$F_n(\xi_{p,n}(x)|x) = p. \quad (8)$$

We define the first partial derivative with respect to the second component of

$F_{1,n}(x, t)$ and $\tilde{F}_{1,n}(x, t)$ respectively by

$$\frac{\partial F_{1,n}(x, t)}{\partial t} = F'_{1,n}(x, t) = \frac{1}{nh_K h_H} \sum_{i=1}^n \frac{\delta_i}{\bar{G}_n(Y_i)} K\left(\frac{x - X_i}{h_K}\right) H'\left(\frac{t - Y_i}{h_H}\right),$$

and

$$\frac{\partial \tilde{F}_{1,n}(x, t)}{\partial t} = \tilde{F}'_{1,n}(x, t) = \frac{1}{nh_K h_H} \sum_{i=1}^n \frac{\delta_i}{\bar{G}_n(Y_i)} K\left(\frac{x - X_i}{h_K}\right) H'\left(\frac{t - Y_i}{h_H}\right)$$

where H' is the derivative of H .

The conditional densities estimators are given by

$$f_n(t|x) = \frac{F'_{1,n}(x, t)}{\ell_n(x)}$$

and

$$\tilde{f}_n(t|x) = \frac{\tilde{F}'_{1,n}(x, t)}{\ell_n(x)} \quad (9)$$

In order to study the behaviour of the random variable $(\xi_p(x) - \xi_{p,n}(x))$, we make use of the properties in (3) and (8) and Taylor's expansion, we get

$$F(\xi_p(x)|x) - F(\xi_{p,n}(x)|x) = (\xi_p(x) - \xi_{p,n}(x)) f(\xi_{p,n}^*(x)|x) \quad (10)$$

where $\xi_{p,n}^*(x)$ lies between $\xi_p(x)$ and $\xi_{p,n}(x)$ and $f(\cdot|x)$ is the conditional density of T given X . It is clear that equation (10), shows that from the behaviour of $(F(\xi_p(x)|x) - F(\xi_{p,n}(x)|x))$, it is easy to obtain asymptotic

results for the sequence $(\xi_p(x) - \xi_{p,n}(x))$. If $f(\xi_{p,n}^*(x)|x) \neq 0$ is not satisfied, we should have to increase the order of Taylor's expansion.

Remark 2. The idea to replace the step function by a smooth one have to pay a price, that is, we have to choose double bandwidths, but as it can be seen in the simulation study, the same choice of the bandwidths does not affect the quality of the estimator and the asymptotic normality.

3. Assumptions and main results

We begin by recalling the definition of the strong mixing property. For this we introduce the following notations. Let $\mathcal{F}_i^k(Z)$ denote the σ -algebra generated by $\{Z_j, i \leq j \leq k\}$.

Definition 1 Let $\{Z_j, i = 1, 2, \dots\}$ be a strictly stationary sequence of random variables. Given a positive integer n , set

$$\alpha(n) = \sup \left\{ |\mathbb{P}(A \cap B) - \mathbb{P}(A)\mathbb{P}(B)| : A \in \mathcal{F}_1^k(Z) \text{ and } B \in \mathcal{F}_{k+n}^\infty(Z), k \in \mathbb{N}^* \right\}.$$

The sequence is said to be α -mixing (strong mixing) if the mixing coefficient $\alpha(n) \rightarrow 0$ as $n \rightarrow \infty$.

There exist many processes fulfilling the strong mixing property. We quote, here, the usual ARMA processes (with innovations satisfying some moment conditions) are geometrically strongly mixing, i.e., there exist $\rho \in (0, 1)$ and $C > 0$ such that, for any $n \geq 1$, $\alpha(n) \leq C\rho_n$ (see, e.g., Jones. 1978).

Let $\Omega_0 = \{x \in \mathbb{R}; \ell(x) > 0\}$, Ω be a compact set such that $\Omega \subset \Omega_0$, $\mathcal{C}$ be a compact set included in $] -\infty, \tau_F[$ and without loss of generality, we assume that $h_K = h_H =: h$. Our assumptions are gathered here together for easy reference.

A1 The kernel K satisfies:

- (i) K is strictly positive valued, bounded, with compact support and there

exist constant M^* , and m^* such that $0 < M^* < \infty$,
 $0 < m^* < \infty$, $\sup_{\mathbb{R}} K(u) = M^*$, and $\inf_{\mathbb{R}} K(u) = m^*$,

- (ii) K is Hölder of order γ for some $\gamma > 1$,
- (iii) $\int_{\mathbb{R}} K(u) du = 0$, $\int_{\mathbb{R}} |u| K(u) du < \infty$ and $\int_{\mathbb{R}} h^2 K(u) du < \infty$,
- (iv) $\kappa = \int_{\mathbb{R}} K^2(u) du < \infty$.

A2 The bandwidth h satisfies:

- (i) $\sqrt{\frac{\log \log n}{n}} = o(h^2)$,
- (ii) $nh^5 \rightarrow 0$ as $n \rightarrow \infty$.

A3 The mixing coefficient α is such that

$$\sum_{k \geq 1} k^\delta (\alpha(k))^{1 - \frac{2}{\nu}} < \infty, \text{ for some } \nu > 2 \text{ and } \delta > 1 - \frac{2}{\nu}.$$

A4 The conditional $dfF(t|x)$ has a positive first derivative with respect to t , for all $x \in \Omega$, denoted $f(t|x)$ and satisfies

- (i) The Lipschitz condition of order η and β with respect to t and x , such that

$$\forall (t_1, t_2) \in \mathbb{R}^2, \forall (x_1, x_2) \in \Omega^2,$$

$$|F(t_1|x_1) - F(t_2|x_2)| \leq C(|x_1 - x_2|^\eta + |t_1 - t_2|^\beta),$$
- (ii) $\int_{\mathbb{R}} |t| f(t|x) dt < \infty$, for all $x \in \Omega$ (that is the conditional density $f(\cdot|x)$ has a finite first moment).

A5 The first derivative H' is positive, Lipschitz and $\mathcal{C}^2$ -probability density such that

- (i) There exist two constants $0 < M < \infty$, and $0 < m < \infty$, $\sup_{\mathbb{R}} H'(t) = M$ and $\inf_{\mathbb{R}} H'(t) = m$,
- (ii) $\int_{\mathbb{R}} t H'(t) dt = 0$ and $\int_{\mathbb{R}} |t|^\beta H'(t) dt < \infty$.

- A6** The marginal density $\ell(\cdot)$ satisfies Lipschitz condition and there exists $\gamma_0 > 0$ such that $\ell(x) > \gamma_0$ for all $x \in \Omega$.
- A7** The function $F_1(\cdot, \cdot)$ is differentiable up to order 3 and $\sup_{x,t} \left| F_1^{(i,j)}(x, t) \right| < \infty$, for $i + j \leq 3$,
 where $F_1^{(i,j)}(x, r) := \frac{\partial^{i+j} F_1(x, t)}{\partial x^i \partial t^3}$.
- A8** The marginal density $\ell_{1,|i-j|}(\cdot, \cdot)$ of (X_i, X_j) is bounded.
- A9** The distribution function of the censored rv G has bounded first derivative g .

Remark 3 Some comments on the assumptions

Assumptions **A1**, **A2** (i), **A4** and **A6** are common in nonparametric estimation. Assumption **A3** is a standard assumption on the decay of a strong mixing coefficient. We note that Assumption **A3** is fulfilled if $\alpha(k) = O(k^{-2(2+\alpha)})$ for $\alpha > 0$. A similar assumption is also employed by Fan & Masry (1992), Masry (2005), Roussas & Tran (1992). Assumption **A4** (i) concerns the Lipschitz condition upon the conditional df which permits us to treat the bias term. Assumption **A7** is needed to deal with the bias term of the joint density $F_1'(\cdot, \cdot)$. Assumption **A8** is used when we deal with the covariance term to make it negligible. **A4** (ii) and **A9** are additional assumptions to get the asymptotic variance term.

Now we are in position to give our main results.

Proposition 1. Under Assumptions **A1–A6**, **A8**, **A9** and for any $x \in \Omega_0$ such that $\ell(x) > 0$, we have

$$(nh)^{1/2} (F_n(t|x) - F(t|x)) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \sigma^2(x, t)) \quad \text{as } n \rightarrow \infty \quad (11)$$

where $\xrightarrow{\mathcal{D}}$ denotes the convergence in distribution, and

$$\sigma^2(x, t) = \frac{\kappa F(t|x) (1 - \overline{G}(t) F(t|x))}{\ell(x) \overline{G}(t)}.$$

Theorem 1 Let $p \in (0, 1)$ such that $p = F(\xi_p(x)|x)$. Under Assumptions **A1–A9**, for any $x \in \Omega_0$ such that $(x, \xi_p(x)) \neq 0$, we have

$$\left(\frac{nh}{\left(\sum (x, \xi_p(x)) \right)^2} \right)^{1/2} (\xi_p(x) - \xi_{p,n}(x)) \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1), \text{ as } n \rightarrow \infty \quad (12)$$

where

$$\left(\sum (x, \xi_p(x)) \right)^2 = \frac{\kappa p (1 - p \overline{G}(\xi_p(x)))}{f^2(\xi_p(x)|x) \ell(x) \overline{G}(\xi_p(x))}.$$

Remark 4 For complete data ($\overline{G}(t) = 1$), Honda (2000) dealt with the α -mixing case and proved the uniform convergence and asymptotic normality of an estimate of the conditional quantile using a polynomial fitting method. Berline et al. (2001) showed the asymptotic normality of convergent kernel conditional quantile estimators by considering the particular case of a stationary α -mixing process. Furthermore we get the same asymptotic variance. Gannoun et al. (2003) gave a smooth nonparametric conditional median predictor, based on double kernel methods and established its asymptotic normality and proposed an extension to the conditional quantile.

It is well known, from the robustness theory that the median is more robust than the mean, therefore the conditional median, $\mu(x) = \xi_{1/2}(x)$, is a good alternative to the conditional mean as a predictor for a variable Y given $X = x$. Note that the estimation of $\mu(x)$ is given by $\mu_n(x) = \xi_{\frac{1}{2},n}(x)$. Using these considerations and Section 2, we want to predict the non observed rv Y_{n+1} (which corresponds to some modality of our problem), from available data $X_1, \dots, X_n$. Given a new value X_{n+1} , we can predict the corresponding response Y_{n+1} by

$$Y_{n+1} = \mu_n(X_{n+1}) = \xi_{\frac{1}{2},n}(X_{n+1}).$$

Applying the above Theorem, we have the following corollary

Corollary 1

$$\left(\frac{nh}{\left(\sum \left(X_{n+1}, \xi_{1/2}(X_{n+1}) \right) \right)^2} \right)^{\frac{1}{2}} \left(\xi_{1/2,n}(X_{n+1}) - \xi_{1/2}(X_{n+1}) \right) \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1), \text{ as } n \rightarrow \infty.$$

Using a plug-in method by replacing $\ell, \bar{G}, f$ and $\xi_{1/2}$ by their estimates $\ell_n, \bar{G}_n, f_n$ and $\xi_{1/2,n}$ respectively, permit us to obtain a convergent estimate $\sum_n$ of $\sum$, then we get from Corollary 1.

Corollary 2

$$\left(\frac{nh}{\left(\sum_n \left(x, \xi_{1/2,n}(x) \right) \right)^2} \right)^{\frac{1}{2}} \left(\xi_{1/2,n}(x) - \xi_{1/2}(x) \right) \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1), \text{ as } n \rightarrow \infty \quad (13)$$

From this corollary, we get for each fixed $\eta \in (0, 1)$, the following approximate $(1 - \eta)\%$ confidence interval

$$\xi_{1/2,n}(x) \pm z_{1-\eta/2} \times \frac{\sum_n \left(x, \xi_{1/2,n}(x) \right)}{\sqrt{nh}}$$

where $z_{1-\eta/2}$ denotes the $1 - \eta/2$ quantile of the standard normal distribution.

4. Simulations

The aim of the following simulations is to examine how the asymptotic approximation performs for finite samples sizes in some particular case. We consider the strong mixing bidimensional process generated by:

$$\begin{cases} X_i = \rho X_{i-1} + \sqrt{1 - \rho^2} \varepsilon_i, \\ T_i = X_{i+1} \quad i = 1, \dots, n \end{cases}$$

where $0 < \rho < 1$, (ε_i) is a white noise sequence and X_0 is a standard normal rv independent of (ε_i) . Clearly, the process (X_n) is an AR(1) and given $X_1 = x$, we have $T_1 = \rho x + \sqrt{1 - \rho^2} \varepsilon_2$, hence it is distributed as a Gaussian rv

$\mathcal{N}(\rho x, 1 - p^2)$. We also simulate n iid rv C_i exponentially distributed with parameter $\lambda = 0.2$. Clearly, the process (T_n, X_n, C_n) is stationary and strong mixing. Then we compute our estimator with the observed data (X_i, Y_i, δ_i) , where $Y_i = T_i \wedge C_i$ and $\delta_i = \mathbb{I}_{\{T_i \leq C_i\}}$, and we use the standard Gaussian kernel. The bandwidth was taken to be $h = 1.97 \left(\frac{\log n}{n} \right)^{\frac{1}{4}}$ which satisfies Assumption **A2**. Here we point out that this choice can be made by cross validation method. We have chosen H such that $H' = K$.

4.1 Asymptotic normality

We show how good the normality is, when dealing with samples of finite size which is the case in practice. The data arise from the same distribution as previously for a given size n . We estimate the conditional quantile function as before and calculate the normalised deviation between this estimate and the theoretical conditional quantile for $x = 0$, $p = 1/2$, $\rho = 0.8$ and $\sigma = 0.5$

$$\bar{\xi}_{p,n}(x) = \sqrt{nh} \left(\frac{\kappa p (1 - p \bar{G}_n(\xi_p(x)))}{f_n^2(\xi_p(x)|x) \ell_n(x) \bar{G}_n(\xi_p(x))} \right) (\xi_{p,n}(x) - \xi_p(x))$$

which becomes

$$\bar{\xi}_{p,n}(0) = \sqrt{nh} \left(\frac{\kappa \frac{1}{2} (1 - \frac{1}{2} \bar{G}_n(0))}{f_n^2(0|0) \ell_n(0) \bar{G}_n(0)} \right)^{-\frac{1}{2}} (\xi_{\frac{1}{2},n}(x) - 0).$$

We draw, using this scheme, B independent n -samples. In order to estimate the density function of $\xi_{1/2,n}(x)$ (by the kernel method), we make the classical bandwidth choice (see, e.g. Silverman, 1986 p. 40) $h' = CB^{-1/5}$, where the constant C is appropriately chosen.

We see that the normality of the rv $\xi_{1/2,n}(\cdot)$ is better as n increases, which is also clear from the Q.Q. plot.

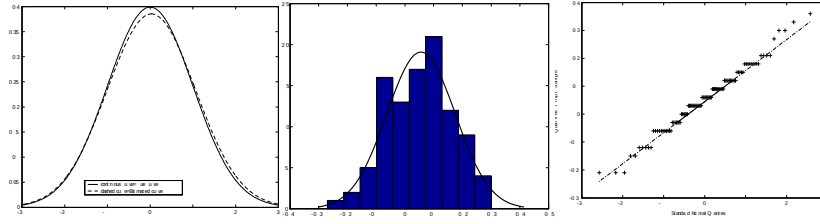


Figure 1. $n = 100$, $B = 100$

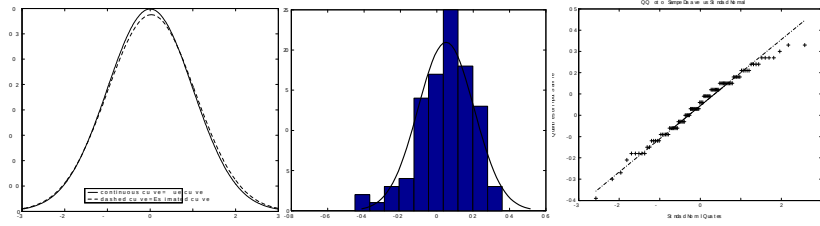


Figure 2. $n = 200$, $B = 100$

4.2 Confidence intervals

We built an approximate $(1 - \eta)\%$ confidence intervals obtained from Corollary 2, that is

$$\xi_{1/2,n} \left(x \pm z_{1-\eta/2} \times \frac{\sum_n \left(x, \xi_{1/2,n}(x) \right)}{\sqrt{nh}} \right)$$

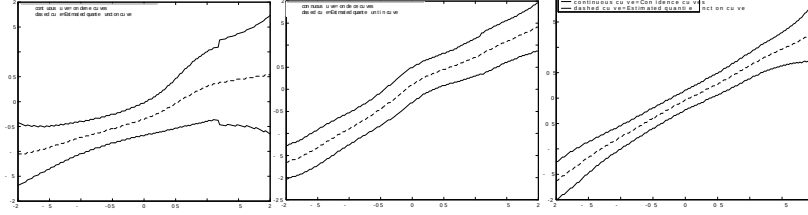


Figure 3. $p = 1/2$; $n = 50$, $n = 100$ and $n = 200$ respectively

We draw in Figure 3, for several values of n , three curves corresponding to the estimator, the approximate $(1 - \eta/2)\%$ lower and upper confidence interval, for $p = 1/2$, $x \in [-2, 2]$, $\sigma = 0.5$ and $\eta = 5\%$. The bandwidth h is chosen as in paragraph 5.1. Figure 3 suggests that approximate confidence intervals perform well for moderately large sample sizes.

5. Auxiliary results and proofs

First Lemma deals with the uniform convergence of the conditional probability density estimator $f_n(t|x)$ to $f(t|x)$.

Lemma 1 Under Assumptions **A1**, **A2** (i), **A3**, **A5–A7**, we have,

$$\sup_{t \in C} |f_n(t|x) - f(t|x)| \rightarrow 0 \quad \text{a.s. as } n \rightarrow \infty. \quad (14)$$

Proof of Lemma 1

$$f_n(t|x) - f(t|x) = \left(f_n(t|x) - \tilde{f}_n(t|x) \right) + \left(\tilde{f}_n(t|x) - f(t|x) \right).$$

$$\begin{aligned}
\left| f_n(t|x) - \tilde{f}_n(t|x) \right| &= \frac{1}{\ell_n(x)} \left| \left(F'_{1,n}(x, t) - \tilde{F}'_{1,n}(x, t) \right) \right| \\
&= \frac{1}{\ell_n(x) n h^2} \left| \sum_{i=1}^n \delta_i K\left(\frac{x - X_i}{h}\right) H'\left(\frac{t - Y_i}{h}\right) \right. \\
&\quad \cdot \left. \left(\frac{1}{\overline{G}_n(Y_i)} - \frac{1}{\overline{G}(Y_i)} \right) \right| \\
&\leq \frac{1}{\ell_n(x)} \frac{1}{h_n^2} \frac{M^* M}{\overline{G}_n(\tau_F) \overline{G}(\tau_F)} \\
&\quad \cdot \sup_{t \in C} |F_n(t) - G(t)| \frac{1}{n} \sum_{i=1}^n \delta_i.
\end{aligned}$$

In the same way as in the Theorem 2 of Cai (2001), it can be shown by A3 that $\sup_{t \in C} |\overline{G}_n(t) - \overline{G}(t)| = O(\log \log n / n)^{\frac{1}{2}}$ a.s.

Since $\ell_n(x) \geq \inf_x \ell_n(x) \geq \gamma_0 > 0$ and $\overline{G}(\tau_F) > 0$, in conjunction with the SLLN (see Thm 2.2 in Cai & Roussas, 1992) and under Assumption **A3**, we have

$$\sup_{t \in C} \left| f_n(t|x) - \tilde{f}_n(t|x) \right| \leq \frac{c M^* M}{\overline{G}^2(\tau_F) \gamma_0 h_n^2} \left(\frac{\log \log n}{n} \right)^{\frac{1}{2}} \quad \text{a.s.}$$

where c is a positive constant. Assumption **A2** (i) gives us that

$$\sup_{t \in C} \left| f_n(t|x) - \tilde{f}_n(t|x) \right| \rightarrow 0 \quad \text{a.s. as } n \rightarrow \infty.$$

Let now examine the difference $\left(\tilde{f}_n(t|x) - f(t|x) \right)$. By formula (9) we have the following inequality.

$$\begin{aligned}
\sup_{t \in C} \left| \tilde{f}_n(t|x) - f(t|x) \right| &\leq \frac{1}{\inf_{x \in \Omega} \ell_n(x)} \left\{ \sup_{t \in C} \left| F'_{1,n}(x, t) - \mathbb{E} \left[\tilde{F}'_{1,n}(x, t) \right] \right| \right. \\
&\quad + \sup_{t \in C} \left| \mathbb{E} \left[\tilde{F}'_{1,n}(x, t) \right] - F_1(x, t) \right| \\
&\quad \left. + |[\ell_n(x) - \ell(x)]| \sup_{t \in C} |f(t|x)| \right\} \\
&=: \frac{1}{\inf_{x \in \Omega} \ell_n(x)} \left\{ \mathcal{L}_1 + \mathcal{L}_2 + \mathcal{L}_3 \sup_{t \in C} |f(t|x)| \right\}. \quad (15)
\end{aligned}$$

Now deal with $\mathcal{L}_1$ and set

$$U_i(x, t) = \frac{1}{nh^2} \frac{\delta_i}{\overline{G}(Y_i)} K\left(\frac{x-X_i}{h}\right) H'\left(\frac{t-Y_i}{h}\right) - \mathbb{E}\left[\frac{1}{nh^2} \frac{\delta_i}{\overline{G}(Y_i)} K\left(\frac{x-X_i}{h}\right) H'\left(\frac{t-Y_i}{h}\right)\right].$$

Clearly

$$\mathcal{L}_1 = \sum_{i=1}^n U_i(x, t).$$

As C is a compact set, it can be covered by finite numbers I_n of intervals centered at t_j of length b_n such that $I_n b_n \leq C$ and for any $t \in C$ there exists a point t_j such that $|t - t_j| \leq b_n$. Moreover,

$$\left| \sum_{i=1}^n U_i(x, t) \right| \leq \left| \sum_{i=1}^n U_i(x, t) - \sum_{i=1}^n U_i(x, t_j) \right| + \left| \sum_{i=1}^n U_i(x, t_j) \right|.$$

From **A2** (i) and the first part of **A5**, it follows that

$$\left| \sum_{i=1}^n U_i(x, t) - \sum_{i=1}^n U_i(x, t_j) \right| \leq \frac{2Mn}{nh^2 \overline{G}(\tau)} \left| \frac{t - t_j}{h} \right| \leq \frac{C}{h^3 I_n}.$$

Hence

$$\mathcal{L}_1 \leq \frac{C}{h^3 I_n} + \max_{1 \leq j \leq I_n} \left| \sum_{i=1}^n U_i(x, t_j) \right|.$$

It suffices to choose $I_n = O(n)$ and to show that $\forall \varepsilon > 0$

$$\mathbb{P}\left(\max_{1 \leq j \leq I_n} \left| \sum_{i=1}^n U_i(x, t_j) \right| > \varepsilon\right) \rightarrow 0 \quad \text{as } n \rightarrow \infty. \quad (16)$$

Clearly, we have

$$\mathbb{P}\left(\max_{1 \leq j \leq I_n} \left| \sum_{i=1}^n U_i(x, t_j) \right| > \varepsilon\right) \leq I_n \max_{1 \leq j \leq I_n} \mathbb{P}\left(\left| \sum_{i=1}^n U_i(x, t_j) \right| > \varepsilon\right).$$

The rv's $U_i(x, t_j)$ are strongly mixing, centered and bounded by $d = \frac{C}{nh^2}$. In order to provide an exponential bound for the last term in the right hand side of the last inequality, we use a version of a lemma due to Bosq (1998) (p. 25–26), that is $\forall n \in \mathbb{N}, \forall i \in \mathbb{N}, 1 \leq i \leq n, |U_i| \leq d$ then for each $q \in \left[1, \frac{n}{2}\right]$ and

for each $\varepsilon > 0$

$$\begin{aligned} \mathbb{P} \left(\left| \sum_{i=1}^n U_i(x, t_j) \right| > n\varepsilon \right) &\leq 4 \exp \left(-\frac{\varepsilon^2 q}{8} \left[\frac{8q^2 \sigma^2(q)}{n^2} + \frac{d\varepsilon}{2} \right]^{-1} \right) \\ &\quad + 22 \left(1 + \frac{4d}{\varepsilon} \right)^{\frac{1}{2}} q \alpha \left(\left[\frac{n}{2q} \right] \right) \\ &=: \mathcal{J}_1 + \mathcal{J}_2 \end{aligned} \quad (17)$$

where $\sigma^2(q) = \frac{K_1 n}{2q} \mathbb{E} [U_1^2(x, t_j)]$.

$$\begin{aligned} &\mathbb{E} (U_1^2(x, t_j)) \\ &= \mathbb{E} \left[\frac{1}{n^2 h^4} \frac{\delta_1}{\overline{G}^2(Y_1)} K^2 \left(\frac{x - X_1}{h} \right) H'^2 \left(\frac{t_j - Y_1}{h} \right) \right] \\ &\quad - \left(\mathbb{E} \left[\frac{1}{n h^2} \frac{\delta_1}{\overline{G}(Y_1)} K \left(\frac{x - X_1}{h} \right) H' \left(\frac{t_j - Y_1}{h} \right) \right] \right)^2 \\ &=: \nu_1 - \nu_2. \\ \nu_1 &= \frac{1}{n^2 h^4} \mathbb{E} \left[\mathbb{E} \left(\frac{\delta_1}{\overline{G}^2(Y_1)} K^2 \left(\frac{x - X_1}{h} \right) H'^2 \left(\frac{t_j - Y_1}{h} \right) \middle| X_1 \right) \right] \\ &= \frac{1}{n^2 h^4} \mathbb{E} \left[\mathbb{E} \left[\left(\frac{\delta_1}{\overline{G}^2(Y_1)} K^2 \left(\frac{x - X_1}{h} \right) H'^2 \left(\frac{t_j - Y_1}{h} \right) \right) \middle| T_1 \right] \middle| X_1 \right]. \end{aligned}$$

Assumptions **A1** (i), **A5** (i) and **A6** give that $\nu_1 \leq \frac{C}{n^2 h^4}$. The same arguments allow us to have $\nu_2 \leq \frac{C}{n^2 h^4}$, so $\mathbb{E} (\Delta_1^2(x, t_j)) = O \left(\frac{1}{n^2 h^4} \right)$.

Now taking $q = C n^\kappa k$ for some $\kappa < 1$ we have, for n large enough, $\frac{q^2 \sigma^2(q)}{n^2}$ is of order $\frac{1}{n^{3-\kappa} h^3}$ and is therefore asymptotically negligible than $\frac{d\varepsilon}{2}$. Then for n large enough

$$\mathcal{J}_1 \leq 4 \exp \left(-C\varepsilon \frac{1}{n^3 h^2} \right).$$

From **A3** the second term in (17) can be bounded by $\mathcal{J}_2 \leq C\varepsilon \frac{h^{\frac{\delta\nu}{\nu-2}}}{n^{(1-\kappa)\frac{\delta\nu}{\nu-2}+1}}$.

Taking $\varepsilon = \varepsilon_0 \frac{1}{n}$ we obtain

$$\mathbb{P} \left(\max_{1 \leq j \leq I_n} \left| \sum_{i=1}^n U_i(x, t_j) \right| > \varepsilon \right) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Then

$$\mathcal{L}_1 \rightarrow 0 \quad \text{a.s. as } n \rightarrow \infty. \quad (18)$$

Now we deal with the bias term $\mathcal{L}_2$ which is independent of the structure of the process. Clearly, we have

$$\begin{aligned} \mathbb{E} \left[\tilde{F}'_{1,n}(x, t) \right] &= \frac{1}{h_n^2} \mathbb{E} \left\{ \mathbb{I}_{\{T_1 \leq C_1\}} \bar{G}^{-1}(Y_1) K \left(\frac{x - X_1}{h} \right) H' \left(\frac{t - Y_1}{h} \right) \right\} \\ &= \frac{1}{h_n^2} \mathbb{E} \left\{ K \left(\frac{x - X_1}{h} \right) \mathbb{E} \left[\mathbb{I}_{\{T_1 \leq C_1\}} \bar{G}^{-1}(Y_1) \right. \right. \\ &\quad \left. \left. H' \left(\frac{t - Y_1}{h} \right) \middle| X_1, T_1 \right] \right\}. \end{aligned}$$

Using $\mathbb{I}_{\{T_1 \leq C_1\}} \varphi(Y_1) = \mathbb{I}_{\{T_1 \leq C_1\}} \varphi(T_1)$, a change of variable and a Taylor expansion, we get, under **A1**, **A5** (ii) and **A7**

$$\begin{aligned} \mathcal{L}_2 &= \frac{1}{h_n^2} \mathbb{E} \left\{ K \left(\frac{x - X_1}{h} \right) H' \left(\frac{t - T_1}{h} \right) \bar{G}^{-1}(T_1) \mathbb{E} \left[\mathbb{I}_{\{T_1 \leq C_1\}} \middle| X_1, T_1 \right] \right\} \\ &\quad - F_1(x, t) \\ &= O(h^2). \end{aligned} \quad (19)$$

Now, for $\mathcal{L}_3$. Under Assumptions **A1**, **A2** (i) and **A6** we get easily that $\mathcal{L}_3 = o(1)$. As the kernel estimator $\ell_n(x)$ is almost surely bounded away from zero on Ω , (18), (19) and the last point permit to achieve the proof of Lemma 1.

Remark 5 Working as in Lemma 1 and under more restrictive conditions upon the bandwidth h (see Ould Saïd & Sadki, 2008b), we can get

$$F_n(t|x) \rightarrow F(t|x) \quad \text{a.s. as } n \rightarrow \infty.$$

In order to prove Proposition 1, we will use the following decomposition and the lemmas below

$$\begin{aligned} \tilde{F}_n(t|x) - F(t|x) &= \frac{\ell(x)}{\ell_n(x)} \\ &\cdot \left[\frac{\tilde{F}_{1,n}(x, t) - \mathbb{E}(\tilde{F}_{1,n}(x, t)) - F(t|x)[\ell_n(x) - \mathbb{E}(\ell_n(x) - \mathbb{E}(\ell_n(x)))]}{\ell(x)} \right] \\ &- \frac{1}{\ell_n(x)} \left[F_1(x, t) - \mathbb{E}(\tilde{F}_{1,n}(x, t)) - F(t|x)[\ell(x) - \mathbb{E}(\ell_n(x))]] \right]. \end{aligned}$$

Note that

$$\begin{aligned} (nh)^{\frac{1}{2}} (F_n(t|x) - F(t|x)) &= \frac{\ell(x)}{\ell_n(x)} \left((nh)^{\frac{1}{2}} A_n(x, t) \right) \\ &+ (nh)^{\frac{1}{2}} C_n(x, t), \end{aligned} \quad (20)$$

where

$$\begin{aligned} &\frac{\tilde{F}_{1,n}(x, t) - \mathbb{E}(\tilde{F}_{1,n}(x, t)) - F(t|x)[\ell_n(x) - \mathbb{E}(\ell_n(x))]}{\ell(x)} \\ B_n(x, t) &= \frac{F_1(x, t) - \mathbb{E}(\tilde{F}_{1,n}(x, t)) - F(t|x)[\ell(x) - \mathbb{E}(\ell_n(x))]}{\ell_n(x)} \end{aligned} \quad (21)$$

and

$$C_n(x, t) = \left(F_n(t|x) - \tilde{F}_n(t|x) \right) - B_n(x, t).$$

The next Lemma show the asymptotic normality of $(nh)^{\frac{1}{2}} A_n(x, y)$ and the convergence in probability of $(nh)^{\frac{1}{2}} C_n(x, y)$ to zero. We begin to show the second part, for this we write that

$$\begin{aligned} (nh)^{\frac{1}{2}} C_n(x, y) &= \frac{(nh)^{\frac{1}{2}}}{\ell_n(x)} \left[\left(F_{1,n}(x, t) - \tilde{F}_{1,n}(x, t) \right) \right. \\ &\quad \left. - \left(F_1(x, t) - \mathbb{E}(\tilde{F}_{1,n}(x, t)) \right) \right] \\ &\quad + \frac{(nh)^{\frac{1}{2}}}{\ell_n(x)} [F(t|x)(\ell(x) - \ell_n(x))] \end{aligned} \quad (22)$$

Lemma 2 Under Assumptions **A1** (i), (ii), **A2**, **A3**, **A4** (i), **A5** and **A6** we have

$$(nh)^{\frac{1}{2}} C_n(x, y) \rightarrow 0 \quad \text{in probability as } n \rightarrow \infty.$$

Proof of Lemma 2

On the one hand, making use of the results of Lemma 1 in Ould Saïd & Sadki (2008b), under Assumptions **A1** (ii), **A4** (i), **A5** and **A6** we have that the second term in brackets (22) goes to zero in probability as n goes to infinity.

For the first term in brackets (22) we have,

$$\begin{aligned} \left| F_{1,n}(t, x) - \tilde{F}_{1,n}(t, x) \right| &\leq \frac{1}{nh} \sum_{i=1}^n \left| \delta_i K\left(\frac{x - X_i}{h}\right) H\left(\frac{t - Y_i}{h}\right) \right| \\ &\quad \left| \frac{1}{\bar{G}_n(Y_i)} - \frac{1}{\bar{G}(Y_i)} \right| \\ &\leq \frac{1}{h} \frac{M^*}{\bar{G}_n(\tau_F) \bar{G}(\tau_F)} \sup_{t \in C} |G_n(t) - G(t)| \frac{1}{n} \sum_{i=1}^n \delta_i. \end{aligned}$$

In conjunction with the SLLN and the LIL on the censoring law (see Cai & Roussas, 1992 and Cai, 2001 respectively), Assumptions **A1** (i), **A2** (i) and **A3** permit us to have

$$\sup_{x \in \Omega} \sup_{t \in C} \left| F_{1,n}(t, x) - \tilde{F}_{1,n}(t, x) \right| \xrightarrow{a.s.} 0 \quad \text{as } n \rightarrow \infty.$$

By **A3** and **A6** ($\ell(x) - \ell_n(x)$) tends to zero in probability as n goes to infinity (see Theorem 3.2 in Cai & Roussas, 1992).

On the other hand,

$$\frac{(nh)^{\frac{1}{2}}}{\ell_n(x)} = \frac{(nh)^{\frac{1}{2}} \widetilde{f}_n(t|x)}{\ell_n(x) \widetilde{f}_n(t|x)} = \frac{(nh)^{\frac{1}{2}} \widetilde{f}_n(t|x)}{\tilde{F}'_{1,n}(x, t)},$$

where

$$\tilde{F}'_{1,n}(x, t) = \frac{1}{nh_n^2} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right) H'\left(\frac{t - T_i}{h}\right) \frac{\delta_i}{\bar{G}(Y_i)}.$$

By Assumptions **A1** (i) and **A5** (i), we obtain that $\tilde{F}'_{1,n}(x, t) \geq \frac{m^* m}{h_n^2} \frac{1}{n} \sum_{i=1}^n \delta_i$.

Therefore

$$\frac{(nh)^{\frac{1}{2}}}{\ell_n(x)} \leq \frac{(nh_n^5)^{\frac{1}{2}}}{m^*m} \frac{\tilde{f}_n(t|x)}{\frac{1}{n} \sum_{i=1}^n \delta_i}.$$

By **A2** (ii), **A3**, the convergence of $\tilde{f}_n(t|x)$ (see the proof of Lemma 1) and in conjunction with the SLLN the right-hand side of the last inequality goes to zero in probability as n goes to infinity, which completes the proof of Lemma 2. $\blacksquare$

Note that (21) can be written as

$$\begin{aligned} A_n(x, t) &= \frac{1}{nh\ell(x)} \sum_{i=1}^n \left[K\left(\frac{x-X_i}{h}\right) H\left(\frac{t-Y_i}{h}\right) \frac{\delta_i}{\overline{G}(Y_i)} \right. \\ &\quad \left. - \mathbb{E} \left[K\left(\frac{x-X_i}{h}\right) H\left(\frac{t-Y_i}{h}\right) \frac{\delta_i}{\overline{G}(Y_i)} \right] \right] \\ &\quad - F(t|x) \sum_{i=1}^n \left[K\left(\frac{x-X_i}{h}\right) - \mathbb{E} \left(K\left(\frac{x-X_i}{h}\right) \right) \right] \\ &=: \frac{1}{nh\ell(x)} \sum_{i=1}^n N_i(x, t) \end{aligned}$$

where

$$N_i(x, t) = K_i \left[H_i \frac{\delta_i}{\overline{G}(Y_i)} - F(t|x) \right] - \mathbb{E} \left[K_i \left(H_i \frac{\delta_i}{\overline{G}(Y_i)} - F(t|x) \right) \right],$$

$$K_i := K\left(\frac{x-X_i}{h}\right) \text{ and } H_i := H\left(\frac{t-Y_i}{h}\right).$$

It follows that

$$\begin{aligned} nh \operatorname{Var}(A_n(x, t)) &= \frac{1}{h\ell^2(x)} \operatorname{Var}(N_1(x, t)) + \frac{1}{nh\ell^2(x)} \\ &\quad \cdot \sum_{|i-j|>0}^n \operatorname{Cov}(N_i(x, t), N_j(x, t)) \\ &=: \sigma_n^2(x, t) + R_n(x, t). \end{aligned} \tag{23}$$

The following lemma will be used to obtain the asymptotic variance term.

Lemma 3 Under Assumptions **A4**, **A5** (ii) and **A9** we have

$$Var \left[\frac{\delta_1}{\overline{G}(Y_1)} H \left(\frac{t - Y_1}{h} \right) | X_1 \right] \rightarrow F(t|x) \left[\frac{1}{\overline{G}(t)} - F(t|x) \right] \quad \text{as } n \rightarrow \infty. \quad (24)$$

Proof of Lemma 3

$$\begin{aligned} Var \left[\frac{\delta_1}{\overline{G}(H_1)} H \left(\frac{t - Y_1}{n} \right) | X_1 \right] &= \mathbb{E} \left[\frac{\delta_1}{\overline{G}^2(Y_1)} H^2 \left(\frac{t - Y_1}{h} \right) | X_1 \right] \\ &\quad - \left\{ \mathbb{E} \left[\frac{\delta_1}{\overline{G}(Y_1)} H \left(\frac{t - Y_1}{h} \right) | X_1 \right] \right\}^2 \\ &=: \mathcal{J}_1 + \mathcal{J}_2. \end{aligned} \quad (25)$$

Using the definition of the conditional expectation, we have by integration by parts and change of variable, we get

$$\begin{aligned} \mathbb{E} \left[\frac{\delta_1}{\overline{G}(Y_1)} H \left(\frac{t - Y_1}{h} \right) | X_1 \right] &= \mathbb{E} \left[\mathbb{E} \left(\frac{\delta_1}{\overline{G}(Y_1)} H \left(\frac{t - Y_1}{h} \right) | T_1 | X_1 \right) \right] \\ &= \int H'(z) (F(t - zh|X_1) - F(t|x)) dz \\ &\quad + \int H'(z) F(t|x) dz. \end{aligned}$$

Under **A4** (i) and **A5** (ii) we get that $\int H'(z) F(t|x) dz \rightarrow F(t|x)$ and $\int H'(z) (F(t - zh|X_1) - F(t|x)) dz = O(h^{\min(\eta, \beta)})$. Therefore the second term in (25) tends to $F^2(t|x)$ as n goes to infinity.

The first term of (25) can be developed as follows

$$\begin{aligned} \mathcal{J}_1 &= \mathbb{E} \left[\mathbb{E} \left(H^2 \left(\frac{t - Y_1}{h} \right) \frac{\delta_1}{\overline{G}^2(Y_1)} | T_1 \right) | X_1 \right] \\ &= \int H^2(z) \frac{1}{\overline{G}(t - hz)} dF(t - hz|X_1). \end{aligned}$$

By Taylor's expansion, we have

$$\begin{aligned}\mathcal{J}_1 &= \int H^2(z) \frac{1}{\overline{G}(t)} dF(t - hz|X_1) \\ &\quad + \frac{h}{\overline{G}^2(t)} \int z H(z) \overline{G}'(t^*) dF(t - hz|X_1) + o(1) \\ &=: \mathcal{J}'_1 + \mathcal{J}'_2\end{aligned}\tag{26}$$

where t^* is between t and $t - hz$.

Under Assumption **A9**, the second term of (26) can be bounded by

$$\mathcal{J}'_2 \leq \frac{h_n^2 \sup_{x \in \mathbb{R}} |g(x)|}{\overline{G}^2(\tau_F)} \int z f(t - hz|X_1) dz,$$

then under Assumption **A4** (ii), we get that $\mathcal{J}'_2 = O(h^2)$.

On the other hand, by integrating by parts we have

$$\begin{aligned}\mathcal{J}'_1 &= \frac{1}{\overline{G}(t)} \int 2H'(z) H(z) (F(t - hz|X_1) - F(t|x)) dz \\ &\quad + \frac{1}{\overline{G}(t)} \int 2H'(z) H(z) F(t|x) dz.\end{aligned}$$

Remark that $\int 2H(z) H'(z) F(t|x) dz = F(t|x)$ and under **A4** (i), we get

$$\int 2H'(z) H(z) (F(t - hz|X_1) - F(t|x)) dz = O(h^{\min(\eta, \beta)})$$

and $\int H^2(z) \frac{1}{\overline{G}(t)} f F(t - hz|X_1) \rightarrow \frac{F(t|x)}{\overline{G}(t)}$ as $n \rightarrow \infty$.

Therefore

$$Var \left[\frac{\delta_1}{\overline{G}(Y_1)} H \left(\frac{t - Y_1}{h} \right) | X_1 \right] \rightarrow F(t|x) \left[\frac{1}{\overline{G}(t)} - F(t|x) \right] \quad \text{as } n \rightarrow \infty.$$

■

Lemma 4 Under Assumptions **A1** (i), **A3**, **A4**, **A5** (i), **A6**, **A8** and **A9** we have

$$\sigma_n^2(x, t) \rightarrow \sigma^2(x, t) \quad \text{as } n \rightarrow \infty, \tag{27}$$

and

$$R_n(x, t) \rightarrow 0 \quad \text{as } n \rightarrow \infty. \quad (28)$$

Proof of Lemma 4

$$\begin{aligned} \sigma_n^2(x, t) &= \frac{1}{h\ell^2(x)} \mathbb{E}(N_1^2(x, t)) \\ &= \frac{1}{h\ell^2(x)} \mathbb{E} \left[K_1^2 \left(H_1 \frac{\delta_1}{\overline{G}(Y_1)} - F(t|x) \right)^2 \right] \\ &\quad - \frac{1}{h\ell^2(x)} \left\{ \mathbb{E} \left[K_1 \left(H_1 \frac{\delta_1}{\overline{G}(Y_1)} - F(t|x) \right) \right] \right\}^2. \end{aligned} \quad (29)$$

On the one hand, we can write the second term in (29) as

$$\begin{aligned} &\frac{1}{h\ell^2(x)} \left\{ \mathbb{E} \left[K_1 \left(H_1 \frac{\delta_1}{\overline{G}(Y_1)} - F(t|x) \right) \right] \right\}^2 \\ &= \frac{h}{\ell^2(x)} \left[\mathbb{E} \left(\tilde{F}_{1,n}(x, t) \right) - F(t|x) \mathbb{E}(\ell_n(x)) \right]. \end{aligned} \quad (30)$$

On the other hand, under assumptions **A1** (i), **A4** (i), **A5** and **A6**, using lemma 1 in Ould Saïd & Sadki (2008b), the last term tends to zero as n goes to infinity.

Now we come back to the first term in (29), write

$$\begin{aligned} &\frac{1}{h\ell^2(x)} \mathbb{E} \left[K_1^2 \left(H_1 \frac{\delta_1}{\overline{G}(Y_1)} - F(t|x) \right)^2 \right] \\ &= \frac{1}{h\ell^2(x)} \mathbb{E} \left[K_1^2 \text{Var} \left(H_1 \frac{\delta_1}{\overline{G}(Y_1)} | X_1 \right) \right] \\ &\quad + \frac{1}{h\ell^2(x)} \mathbb{E} \left[K_1^2 \left(\mathbb{E} \left(H_1 \frac{\delta_1}{\overline{G}(Y_1)} | X_1 \right) - F(t|x) \right)^2 \right]. \end{aligned}$$

Using the same arguments as in Lemma 3, we have

$$\frac{1}{h\ell^2(x)} \mathbb{E} \left[K_1^2 \left(\mathbb{E} \left(H_1 \frac{\delta_1}{\overline{G}(Y_1)} | X_1 \right) - F(t|x) \right)^2 \right] \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

under Assumptions **A4**, **A5** (ii) and **A9**, Lemma 3 gives

$$\frac{1}{h\ell^2(x)} \mathbb{E} \left[K_1^2 \text{Var} \left(H_1 \frac{\delta_1}{\overline{G}(Y_1)} | X_1 \right) \right] \rightarrow \sigma^2(x, t).$$

To prove (28), let us split the sum as follows

$$\begin{aligned} R_n(x, t) &= \frac{1}{nh\ell^2(x)} \sum_{0 \leq |i-j| \leq c_n} \text{Cov}(N_i(x, t), N_j(x, t)) \\ &\quad + \frac{1}{nh\ell^2(x)} \sum_{c \leq |i-j| < n} \text{Cov}(N_i(x, t), N_j(x, t)) \\ &=: \mathcal{L}_1 + \mathcal{L}_2 \end{aligned}$$

where $c_n = o(n)$ at a rate specified as below.

For $\mathcal{L}_1$ we have

$$\begin{aligned} \text{Cov}(N_i(x, t), N_j(x, t)) &= \mathbb{E}(N_i(x, t) N_j(x, t)) \\ &\leq \mathbb{E} \left[K_i K_j \left(H_i \frac{\delta_i}{\overline{G}(Y_i)} - F(t|x) \right) \right. \\ &\quad \left. \left(H_j \frac{\delta_j}{\overline{G}(Y_j)} - F(t|x) \right) \right] \\ &\quad + \left[\mathbb{E} \left(K_1 \left(H_1 \frac{\delta_1}{\overline{G}(Y_1)} - F(t|x) \right) \right) \right]^2. \end{aligned}$$

We have $\left| H_i \frac{\delta_i}{\overline{G}(Y_i)} - F(t|x) \right| \leq \frac{1}{\overline{G}(\tau_F)} + 1$, it follows that

$$\text{Cov}(N_i(x, t), N_j(x, t)) \leq \left(\frac{1}{\overline{G}(\tau_F)} + 1 \right)^2 \mathbb{E}(K_i K_j) + o(1) \mathbb{E}(K_1)^2.$$

By **A8** and by **A1** (i) we get $\mathbb{E}(K_i K_j) \leq Ch^2$ and $\mathbb{E}(K_1) \leq Ch$ respectively.

Therefore

$$\text{Cov}(N_i(x, t), N_j(x, t)) \leq C \left(\frac{1}{\overline{G}(\tau_F)} + 1 \right)^2 h^2 + o(1) Ch^2$$

and

$$\mathcal{L}_1 \leq C \frac{hc_n}{\ell^2(x)} + o(1) C \frac{hc_n}{\ell^2(x)}.$$

Now choosing c_n such that hc_n tends to zero when n goes to infinity then we get $\mathcal{L}_1 = o(1)$.

Now for $\mathcal{L}_2$, let us use a version of the moment inequality due to Rio (2000) (p. 588–590). Let p, q, r be real numbers greater than 1 such that $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1$ and $\frac{1}{p} \leq 1 - \frac{2}{\nu}$.

$$\begin{aligned} \mathcal{L}_2 &\leq \frac{1}{nh\ell^2(x)} \sum_{c_n < |i-j| \leq n} C [\mathbb{E}(K_i^q)]^{\frac{1}{q}} [\mathbb{E}(K_j^r)]^{\frac{1}{r}} (\alpha(|i-j|))^{\frac{1}{p}} \\ &\leq \frac{C}{nh\ell^2(x)} \sum_{c_n < |i-j| \leq n} h^{1-\frac{1}{p}} \left(\frac{|i-j|}{c_n} \right)^\delta (\alpha(|i-j|))^{\frac{1}{p}} \\ &\leq C \frac{1}{h^{\frac{1}{p}} c_n^\delta c_n^\delta \ell^2(x)} \sum_{c_n}^\infty k^\delta (\alpha(k))^{1-\frac{1}{p}} \end{aligned}$$

Now choosing p such that $\frac{2}{\nu} < \frac{1}{p}$ and $c_n = O\left(h^{-\frac{2}{p\delta}}\right)$ and by **A3**, we show clearly that $\mathcal{L}_2 \rightarrow 0$ as $n \rightarrow \infty$, which proves (28).

Lemma 5 Under Assumptions **A1** (i), (ii), **A3**, **A4**, **A5** (i), **A6**, **A8** and **A9** we have

$$(nh)^{\frac{1}{2}} A_n(x, t) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \sigma^2(x, t)) \quad \text{as } n \rightarrow \infty \quad (31)$$

where

$$\sigma^2(x, t) = \frac{\kappa F(t|x) (1 - \overline{G}(t) F(t|x))}{\ell(x) \overline{G}(t)}$$

Proof of Lemma 5

We introduce the following notation (to normalise N_i), $\tilde{N}_i = \frac{N_i}{h^{\frac{1}{2}} \ell(x)}$ so that

$$\text{Var}(\tilde{N}_i) = \frac{\text{Var}(N_i)}{h\ell^2(x)} = \sigma_n^2(x, t),$$

under the assumptions of Lemma 4 we have by (27),

$$\text{Var}(\tilde{N}_i) \rightarrow \sigma^2(x, t) \quad \text{as } n \rightarrow \infty$$

and also by (23) and (28)

$$\begin{aligned} \sum_{|i-j|>0} \sum_{i=1}^n \text{Cov} \left(\tilde{N}_i(x, t), \tilde{N}_j(x, t) \right) &= \frac{1}{h\ell^2(x)} \sum_{|i-j|>0} \sum_{i=1}^n \\ \text{Cov} (N_i(x, t), N_j(x, t)) &= o(n). \end{aligned} \quad (32)$$

Now we can write $(nh)^{\frac{1}{2}} A_n(x, t) = \frac{1}{\sqrt{n}} \tilde{S}_n$, where $\tilde{S}_n = \sum_{i=1}^n \tilde{N}_i(x, t)$.

We may show that

$$\frac{1}{\sqrt{n}} \tilde{S}_n \xrightarrow{\mathcal{D}} \mathcal{N}(0, \sigma^2(x, t)) \quad \text{as } n \rightarrow \infty \quad (33)$$

In order to establish the asymptotic normality S_n , we use the Doob technique (see Doob, 1953 pp.228–232), that is the sum $\sum_{i=1}^n \tilde{N}_i(x, t)$ will split up as follows.

Let $0 < M_n < n$, $0 < m_n < n$ be integers tending to infinity along with n and let $0 < k = k_n$ be defined by $k = \left\lfloor \frac{n}{M_n + m_n} \right\rfloor$ (where $[n]$ is the integer part of n) so that $k(M_n + m_n) \leq n$ and $\frac{k(M_n + m_n)}{n} \rightarrow 1$. For $m = 1, \dots, k$, split the set $\{1, 2, \dots, n\}$ into k (large) M_n blocks, I_j and k (small) m_n blocks, J_j as follows

$$I_j = \{i; i = (j-1)(M_n + m_n) + 1, \dots, (j-1)(M_n + m_n) + M_n\},$$

$$J_j = \{i; i = (j-1)(M_n + m_n) + M_n + 1, \dots, j(M_n + m_n)\},$$

the remaining points form the set $\{l; k(M_n + m_n) + 1 \leq l \leq n\}$ (which may be empty).

For $m = 1, \dots, k$ set

$$\begin{aligned} W_j &= \sum_{i=(j-1)(M_n+m_n)+1}^{(j-1)(M_n+m_n)+M_n} \tilde{N}_i, \quad W'_j = \sum_{i=(j-1)(M_n+m_n)+M_n+1}^{j(M_n+m_n)} \tilde{N}_i \\ \text{and } W''_k &= \sum_{i=k(M_n+m_n)+1}^n \tilde{N}_i. \end{aligned}$$

Also set

$$S_{1n} = \sum_{j=1}^k W_j, \quad S_{2n} = \sum_{j=1}^k W'_j, \quad S_{3n} = W''_k, \quad (34)$$

so that $\tilde{S}_n = S_{1n} + S_{2n} + S_{3n}$. The proof of (31) will be obtained by showing

$$\frac{1}{\sqrt{n}} S_{1n} \xrightarrow{\mathcal{D}} \mathcal{N}(0, \sigma^2(x, t)) \quad \text{as } n \rightarrow \infty, \quad (35)$$

and

$$\frac{1}{n} \mathbb{E}(S_{2n}^2) + \frac{1}{n} \mathbb{E}(S_{3n}^2) \rightarrow 0 \quad \text{as } n \rightarrow \infty. \quad (36)$$

Let us prove (36)

$$\begin{aligned} \mathbb{E}(S_{2n}^2) &= \mathbb{E} \left[\left(\sum_{j=1}^k W'_j \right)^2 \right] \\ &= \sum_{j=1}^k \text{Var}[W'_j] + \sum_{|j-l|>0} \sum_{l=1}^k \text{Cov}(W'_j, W'_l) \\ &=: \mathcal{R}_1 + \mathcal{R}_2. \end{aligned}$$

On the one hand, by stationarity and (32) we get

$$\begin{aligned} \text{Var}[W'_j] &= \text{Var} \left[\sum_{i=(j-1)(M_n+m_n)+M_n+1}^{j(M_n+m_n)} \tilde{N}_i \right] \\ &= m_n \text{Var}(\tilde{N}_i) + \sum_{|i-j|>0} \sum_{i=1}^{m_n} \text{Cov}(\tilde{N}_i, \tilde{N}_j) \\ &= m_n \text{Var}(\tilde{N}_i) + o(m_n). \end{aligned} \quad (37)$$

Thus

$$\mathcal{R}_1 = \sum_{j=1}^k \text{Var}[W'_j] = k m_n \text{Var}(\tilde{N}_i) + k o(n).$$

On the other hand, we choose M_n and m_n such as

$$\frac{m_n}{M_n} \rightarrow 0, \quad \frac{M_n}{n} \rightarrow 0; \quad \text{and} \quad \frac{M_n}{\sqrt{nh}} \rightarrow 0, \quad (38)$$

then

$$km_n = \left\lfloor \frac{n}{M_n + m_n} \right\rfloor m_n \approx \frac{nm_n}{M_n + m_n} = o(n),$$

this shows that $\mathcal{R}_1 = o(n)$.

Remark 6 Conditions (38) are easily satisfied, if we choose M_n and m_n as follows $M_n = \lceil n^{\delta_1} \rceil$ and $m_n = \lceil n^{\delta_2} \rceil$ with $0 < \delta^2 < \delta_1 < \frac{1}{3}$.

Now, consider $\mathcal{R}_2$, by stationarity we have

$$\begin{aligned} \mathcal{R}_2 &= \sum_{m=1}^k \sum_{l=1, |m-l|>0}^k Cov(W'_l, W'_m) \\ &= \sum_{|m-l|>0}^k \left(\sum_{i=(m-1)(M_n+m_n)+M_n+1}^{m(M_n+m_n)} \sum_{j=(l-1)(M_n+m_n)+M_n+1}^{l(M_n+m_n)} \right. \\ &\quad \cdot Cov(\tilde{N}_i, \tilde{N}_j) \Big) \\ &\leq \sum_{|m-l|>M_n}^n Cov(\tilde{N}_m, \tilde{N}_l) \\ &\leq o(n). \end{aligned}$$

Finally $\frac{1}{n} \mathbb{E}(S_{2n}^2) \rightarrow 0$ as $n \rightarrow \infty$. Now we deal with the second part of (36), by stationarity we have

$$\begin{aligned} \frac{1}{n} \mathbb{E}(S_{3n}^2) &= \frac{1}{n} Var(S_{3n}^2) \\ &= \frac{1}{n} \left| \sum_{i=k(M_n+m_n)+1}^n Var(\tilde{N}_i) + \sum_{k(M_n+m_n)+1 < |i-j|}^n \right. \\ &\quad \cdot Cov(\tilde{N}_i, \tilde{N}_j) \Big| \\ &\leq \frac{M_n}{n} Var(\tilde{N}_1) + \frac{1}{n} \sum_{|i-j|>0}^n |Cov(\tilde{N}_i, \tilde{N}_j)| \\ &\leq \frac{M_n}{n} Var(\tilde{N}_1) + \frac{o(n)}{n}, \end{aligned}$$

which gives (36).

Now we have to establish (35), this is done in two steps. First, it will be shown that the characteristic functions of $\frac{1}{\sqrt{n}}S_{1n}$ minus the product of the characteristic functions of W_j , $j = 1, \dots, k$ converges to 0 (in absolute value); this proves that the rv W_j are asymptotically independent. Secondly, it is proved that the distribution determined by the product of the characteristic function of W_j , $j = 1, \dots, k$ is asymptotically, the distribution $\mathcal{N}(0, \sigma^2(x, t))$.

The details are as follows; for the first step, we need to use the lemma of Volonskii and Rosanov (see Appendix in Masry, 2005). Note that W_j is $\mathcal{F}_{i_j}^{k_j}$ measurable with $k_j = (j-1)(M_n + m_n) + M_n$ and $i_j = (j-1)(M_n + m_n) + 1$. Then we have

$$\left| \mathbb{E} \left(\exp \left(itn^{-\frac{1}{2}} S_{1n} \right) \right) - \prod_{j=1}^k \mathbb{E} \left(\exp \left(itn^{-\frac{1}{2}} W_j \right) \right) \right| \leq 16k\alpha(m_n + 1).$$

Since $m_n = o(n)$, with an appropriate choice the last term tends to zero as n goes to infinity.

Remark 7 We can take for example m_n as in Remark 6 with $\frac{1}{\nu+1} < \delta^2 < \frac{1}{3}$. Now we have to show that the standard Lindeberg-Feller conditions (see Loève, 1963 p. 280) for asymptotic normality of S_{1n} under independence are satisfied.

So we have to establish

$$\frac{1}{n} \sum_{i=1}^k \mathbb{E} [W_n^2] \rightarrow \sigma^2(x, t) \quad (39)$$

and

$$\frac{1}{n} \sum_{i=1}^k \mathbb{E} \left[W_n^2 \mathbb{I}_{\{|W_i| > \varepsilon \sigma(x, t) \sqrt{n}\}} \right] \rightarrow 0, \quad (40)$$

for all $\varepsilon > 0$.

Working as in (37) with replacing M_n by m_n , we obtain

$$\text{Var}(W_m) = M_n \text{Var}(\tilde{N}_1) + o(M_n)$$

then

$$\frac{1}{n} \sum_{i=1}^k \mathbb{E} [W_n^2] = \frac{kM_n}{n} \text{Var} \left(\tilde{N}_1 \right) + \frac{k}{n} o(M_n).$$

Since $\text{Var} \left(\tilde{N}_1 \right) \rightarrow \sigma^2(x, t)$ and $\frac{M_n}{n} \rightarrow 1$, we get (39).

To prove (40), it suffices to show that, for every $\varepsilon > 0$, the set $\{|W_i| > \varepsilon \sigma(x, t) \sqrt{n}\}$ is empty.

From Assumptions **A1** (i), **A5** (i) and **A6**, there exists a constant C such as

$$\begin{aligned} \tilde{N}_i &= \frac{N_i}{h^{\frac{1}{2}} \ell(x)} \\ &= \frac{1}{h^{\frac{1}{2}} \ell(x)} \left\{ K_i \left[H_i \frac{\delta_i}{\bar{G}(Y_i)} - F(t|x) \right] - \mathbb{E} \left[K_i \left(H_i \frac{\delta_i}{\bar{G}(Y_i)} - F(t|x) \right) \right] \right\} \\ &\leq \frac{C}{h^{\frac{1}{2}}}. \end{aligned}$$

This implies that

$$\frac{1}{\sqrt{n}} |W_m| \leq \frac{M_n C}{\sqrt{nh}}.$$

With the choice of M_n , the right-hand side term of the last inequality goes to zero. So, for n large enough, the set $\{|W_i| > \varepsilon \sigma(x, t) \sqrt{n}\}$ is empty. Then this completes the proof of Lemma 5. $\blacksquare$

The proof of Proposition 1 is, directly completed from Lemma 5.

Proof of Theorem 1

Let $x \in \Omega$. As $F(\cdot|x)$ and $F_n(\cdot|x)$ are continuous, we have $F(\xi_p(x)|x) = F_n(\xi_{p,n}(x)|x) = p$. Then

$$\begin{aligned} |F(\xi_{p,n}(x)|x) - F(\xi_p(x)|x)| &= |F(\xi_{p,n}(x)|x) - F_n(\xi_{p,n}(x)|x)| \\ &\leq \sup_{t \in C} |F_n(t|x) - F(t|x)|. \end{aligned} \quad (41)$$

The convergence in probability of $\xi_{n,p}(x)$ to $\xi_p(x)$ follows immediately from Remark 5 and the continuity of $F(\cdot|x)$.

Using a Taylor expansion we have

$$\xi_{p,n}(x) - \xi_p(x) = \frac{F_n(\xi_{p,n}(x)|x) - F_n(\xi_p(x)|x)}{f_n(\xi_{p,n}^*(x)|x)}$$

where $\xi_{p,n}^*(x)$ lies between $\xi_p(x)$ and $\xi_{p,n}(x)$.

The continuity of $f(\cdot|x)$, Lemma 1 imply the convergence in probability of the above denominator to $f(\xi_p(x)|x)$. Proposition 1 is used to finish the proof. ■

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